

**UNIVERSITY OF RWANDA**  
**COLLEGE OF SCIENCE AND TECHNOLOGY**  
**SCHOOL OF ICT**  
**COMPUTER AND SOFTWARE ENGINEERING DEPARTMENT — NYARUGENGE**  
**CAMPUS**

**MODULE: WEB DESIGN**  
**PROJECT TECHNICAL REPORT**

**ASSESSMENT 2025–2026**

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**CASE STUDY:**

**e-LinkAcre**

*“We Link, Others Follow”*

*A web platform connecting Farmers, Government and Buyers*

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Prepared by: Rwema Eulade  
Registration Number: 225032363  
Lecturer: Dr Eric HITIMANA  
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## Table of Contents

|                                                                            |                                     |
|----------------------------------------------------------------------------|-------------------------------------|
| Table of Contents .....                                                    | 2                                   |
| Table of Figures .....                                                     | 4                                   |
| List of Abbreviations .....                                                | 4                                   |
| Chapter 1. Introduction .....                                              | 5                                   |
| 1.1. Historical Background of the Case Study .....                         | 5                                   |
| 1.2. Problem Statement .....                                               | 5                                   |
| 1.3. Objectives of the Project .....                                       | 5                                   |
| <b>General Objective</b> .....                                             | 5                                   |
| <b>Specific Objectives</b> .....                                           | 5                                   |
| 1.4. Expected Outcomes .....                                               | 6                                   |
| 1.5. Project Rationale (Importance to the Community) .....                 | 6                                   |
| Chapter 2. System Analysis and Design .....                                | 6                                   |
| 2.1. System Analysis .....                                                 | 6                                   |
| <b>2.1.1. Intended Users of the Project</b> .....                          | 6                                   |
| <b>2.1.2. Intended System Partners</b> .....                               | 6                                   |
| 2.2. System Design .....                                                   | 7                                   |
| <b>2.2.1. High-Level Architecture</b> .....                                | 7                                   |
| <b>2.2.2. UI Design (Sketch of the UI Module) and Site Structure</b> ..... | 8                                   |
| <b>2.2.3. Database Design with ERD and Relationships</b> .....             | 9                                   |
| Chapter 3. Implementation .....                                            | 10                                  |
| 3.1. Introduction of the Section .....                                     | 10                                  |
| 3.2. Home Page (index.html) .....                                          | 10                                  |
| 3.3. Remaining Portal Pages .....                                          | <b>Error! Bookmark not defined.</b> |
| Chapter 4. Conclusion and Recommendation .....                             | 14                                  |
| 4.1. State of the Implemented System .....                                 | 14                                  |
| 4.2. Recommendations .....                                                 | 14                                  |
| <b>To the Lecturer, on Technicalities</b> .....                            | 14                                  |
| <b>To the School, on Time Constraints</b> .....                            | 14                                  |
| <b>To the Partners (RAB, NAEB, District Agronomist Offices)</b> .....      | 14                                  |
| <b>To the Users (Farmers, Government Officials, Buyers)</b> .....          | 14                                  |
| <b>To the Beneficiaries (Farming Communities and Buyers)</b> .....         | 14                                  |

|               |    |
|---------------|----|
| Appendix..... | 15 |
|---------------|----|

## Table of Figures

**High-level system architecture of e-LinkAcre..... Error! Bookmark not defined.**  
**Wireframe of the e-LinkAcre home page, reconstructed from index.html... Error! Bookmark not defined.**  
**Harvest update and district-agronomist approval flow ..... Error! Bookmark not defined.**  
**Proposed entity relationship diagram (ERD) for e-LinkAcre ... Error! Bookmark not defined.**

## List of Abbreviations

| Abbreviation | Meaning                                                   |
|--------------|-----------------------------------------------------------|
| <b>ERD</b>   | Entity Relationship Diagram                               |
| <b>GDP</b>   | Gross Domestic Product                                    |
| <b>ICT</b>   | Information and Communication Technology                  |
| <b>NAEB</b>  | National Agricultural Export Development Board            |
| <b>RAB</b>   | Rwanda Agriculture and Animal Resources Development Board |
| <b>UI</b>    | User Interface                                            |
| <b>UI/UX</b> | User Interface / User Experience                          |
| <b>HTML</b>  | HyperText Markup Language                                 |
| <b>CSS</b>   | Cascading Style Sheets                                    |
| <b>JS</b>    | JavaScript                                                |
| <b>MT</b>    | Metric Tonnes                                             |
| <b>CRUD</b>  | Create, Read, Update, Delete                              |

## Chapter 1. Introduction

### 1.1. Historical Background of the Case Study

Agriculture remains the backbone of Rwanda's economy, contributing significantly to national GDP, export earnings and rural livelihoods. Government institutions such as the Rwanda Agriculture and Animal Resources Development Board (RAB) and the National Agricultural Export Development Board (NAEB), together with district agronomists, are responsible for monitoring production, approving harvest data and supporting farmers with extension services and inputs.

Historically, the interaction between farmers, government offices and buyers has been fragmented. Farmers have depended heavily on intermediaries to reach markets, government agencies have lacked a centralized, real-time source of verified harvest data, and buyers have had limited visibility into what is actually being produced, where, and in what quantity. e-LinkAcre project was conceived as a web-based platform to close this gap by bringing Farmers, Government and Buyers onto a single, transparent digital space.

### 1.2. Problem Statement

Farmers in Rwanda frequently sell their productions through middlemen, which reduces the price they receive and limits their direct access to reliable buyers. At the same time, government institutions have no unified digital channel through which to collect and verify harvest statistics at district level, which weakens planning, subsidy allocation and export forecasting. Buyers, in turn, lack a trusted source of verified information, making sourcing decisions slower and less reliable.

There is therefore a need for a single platform that connects these three stakeholders ; farmers, government and buyers around verified, transparent, and up-to-date agricultural production data.

### 1.3. Objectives of the Project

#### **General Objective**

To design and develop e-LinkAcre, a web-based platform that connects Farmers, Government and Buyers in one transparent space, strengthening agriculture as a driver of national development.

#### **Specific Objectives**

- To provide farmers with fair, direct market access to buyers, removing reliance on middlemen.
- To give government institutions (RAB, NAEB, district agronomists) a channel for real-time production reporting to support policy, subsidy planning and food-security forecasting.
- To give buyers a reliable, verified view of available and upcoming harvests for sourcing and export planning.
- To build in district-agronomist approval of harvest data, so that the statistics used by all parties are accountable and trustworthy.
- To design a portal-based structure (Farmer, Government, Buyer, plus public Home/About/Updates/Login pages) that serves each stakeholder group with relevant tools.

## 1.4. Expected Outcomes

- A functioning multi-portal in website (Home, Farmer, Government, Buyer, Updates) implemented in HTML, CSS and JavaScript.
- Farmers are able to reach buyers directly and receive fairer prices based on real market demand.
- Government is able to track agriculture's contribution to GDP and identify exportable surplus using accurate, agronomist-approved harvest data.
- Buyers are able to access verified, forecastable supply information for planning and trade.
- A transparent record of district-agronomist approvals that improves accountability and reduces fraud.

## 1.5. Project Rationale (Importance to the Community)

Agriculture directly affects the livelihoods of the majority of Rwanda's population, and its performance is closely tied to national economic growth, food security and export revenue. By connecting farmers, government and buyers on one platform, e-LinkAcre contributes to fairer trade for farmers, better-informed policy and planning for government, and more reliable sourcing for buyers — supporting the wider national goal of using agriculture as a driver of sustainable development.

# Chapter 2. System Analysis and Design

## 2.1. System Analysis

### 2.1.1. Intended Users of the Project

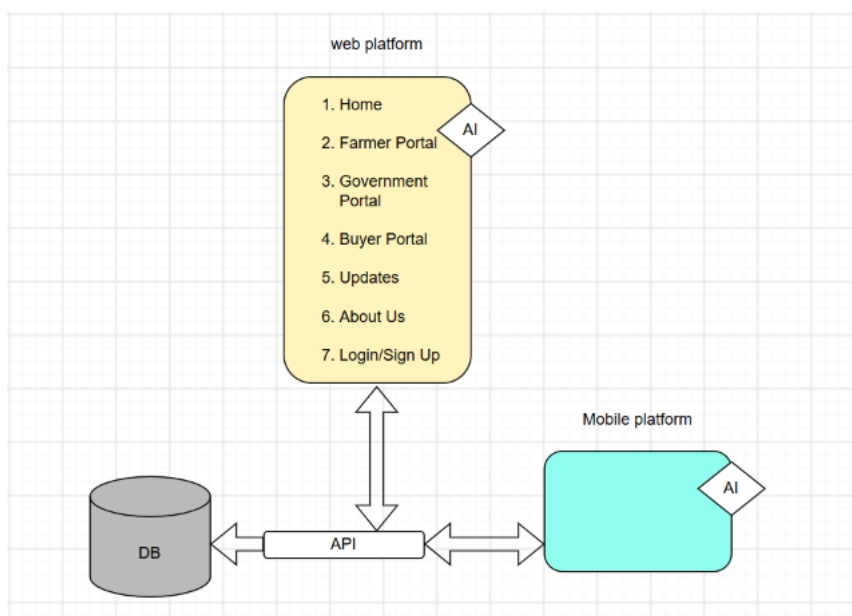
|                                                    |                                                                                                                    |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Farmers</b>                                     | Register, share crop and harvest information, and connect with buyers and government support services.             |
| <b>Buyers</b>                                      | Search verified harvest data, connect directly with farmers, and identify surplus for local or export trade.       |
| <b>Government Officials / District Agronomists</b> | Review and approve harvest submissions, monitor production, and use platform data for policy and subsidy planning. |
| <b>RAB &amp; NAEB Staff</b>                        | Use platform-generated statistics to track agriculture's contribution to GDP and exports.                          |
| <b>e-LinkAcre Agents</b>                           | Support onboarding and coordination between the three stakeholder groups at local level.                           |
| <b>General Public / Visitors</b>                   | View the public Home, About Us and Updates pages before creating an account.                                       |

### 2.1.2. Intended System Partners

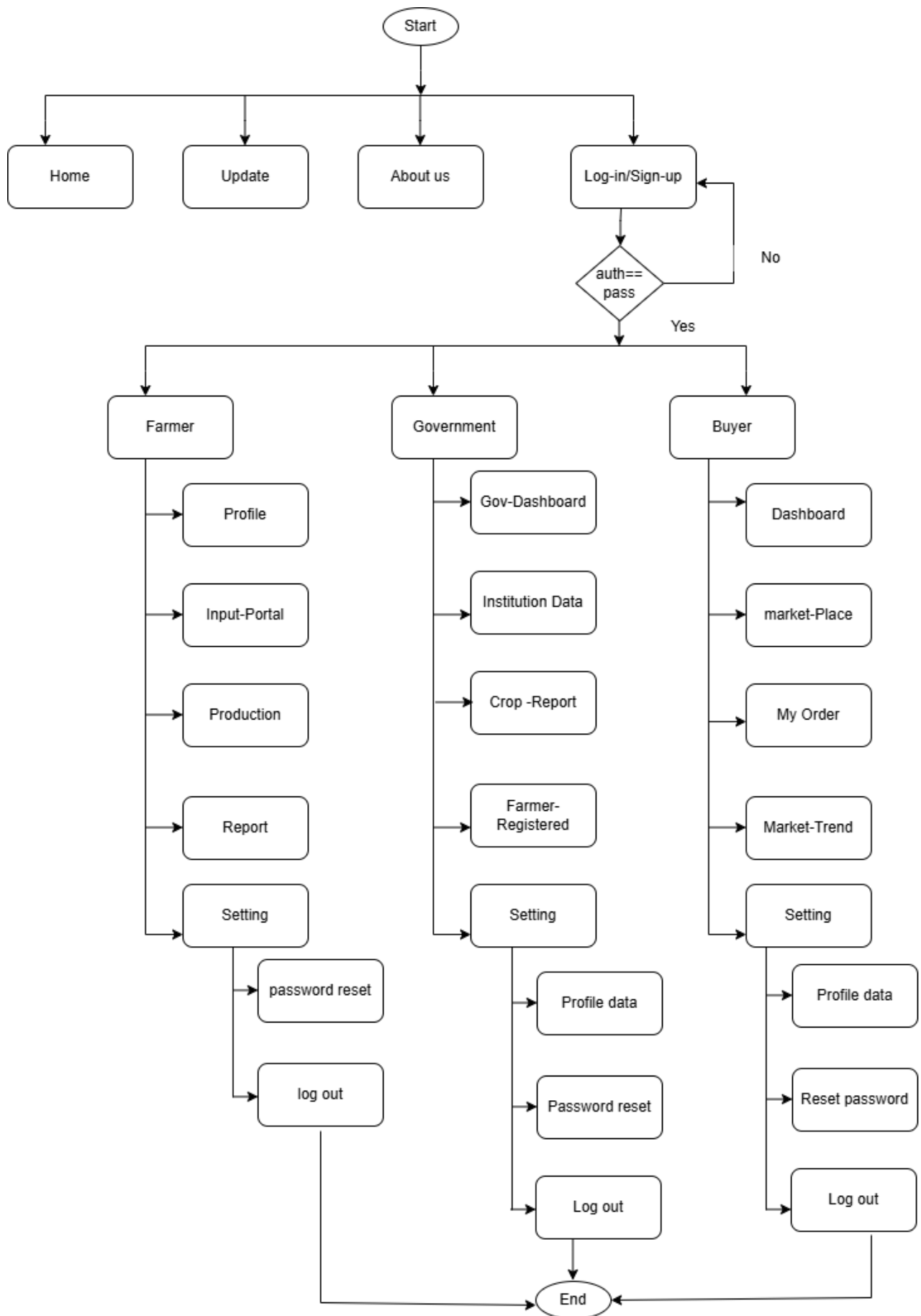
|                                                                        |                                                                                                                      |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Rwanda Agriculture and Animal Resources Development Board (RAB)</b> | Government partner for agricultural production data and technical support.                                           |
| <b>National Agricultural Export Development Board (NAEB)</b>           | Government partner for export-oriented crop data and trade facilitation.                                             |
| <b>District Agronomist Offices</b>                                     | Local government partners are responsible for verifying and approving harvest submissions before they are published. |
| <b>Local Buyers &amp; Exporters</b>                                    | Private-sector partners who consume verified harvest data to plan purchases and exports.                             |

## 2.2. System Design

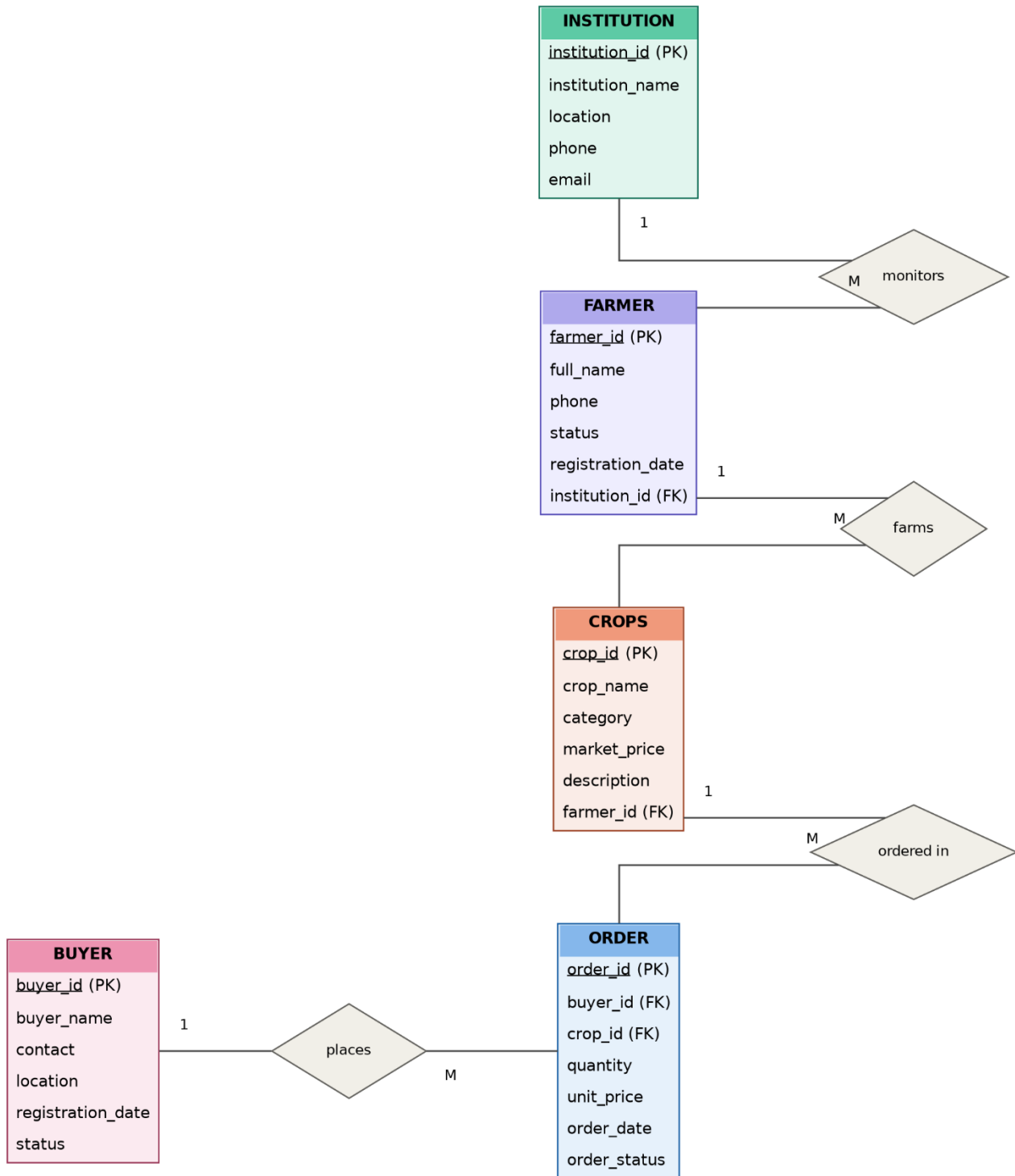
### 2.2.1. High-Level Architecture



### 2.2.2. UI Design (Sketch of the UI Module) and Site Structure



### 2.2.3. Database Design with ERD and Relationship

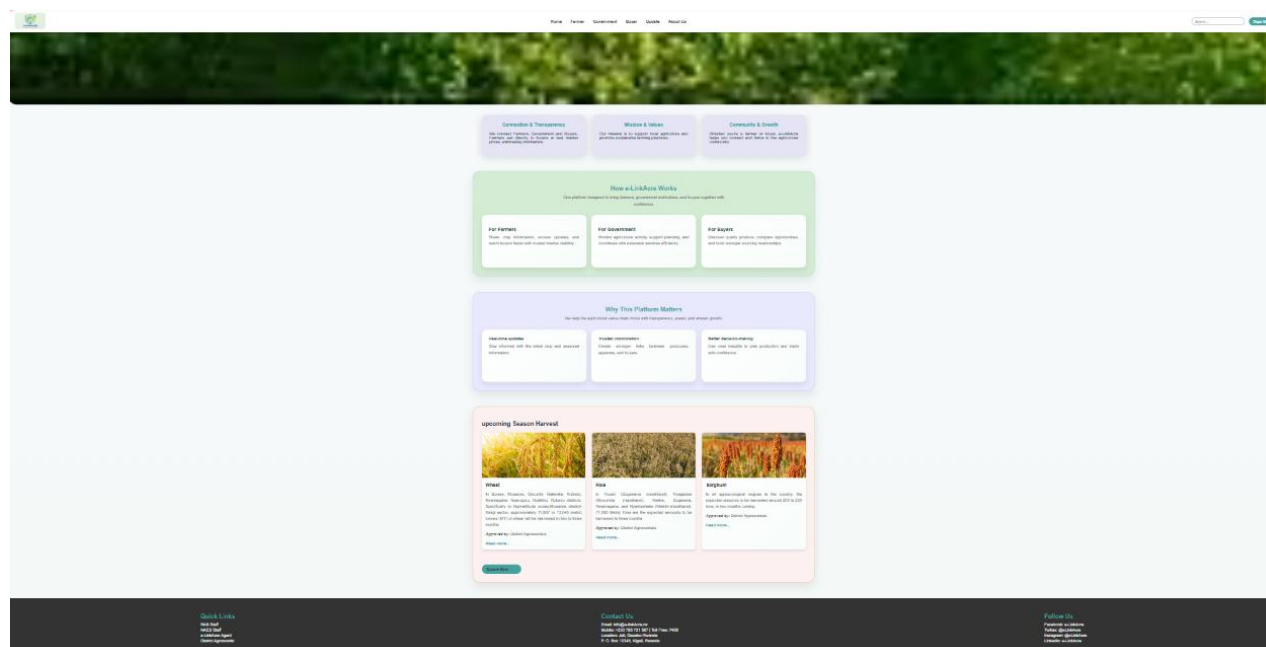


## Chapter 3. Implementation

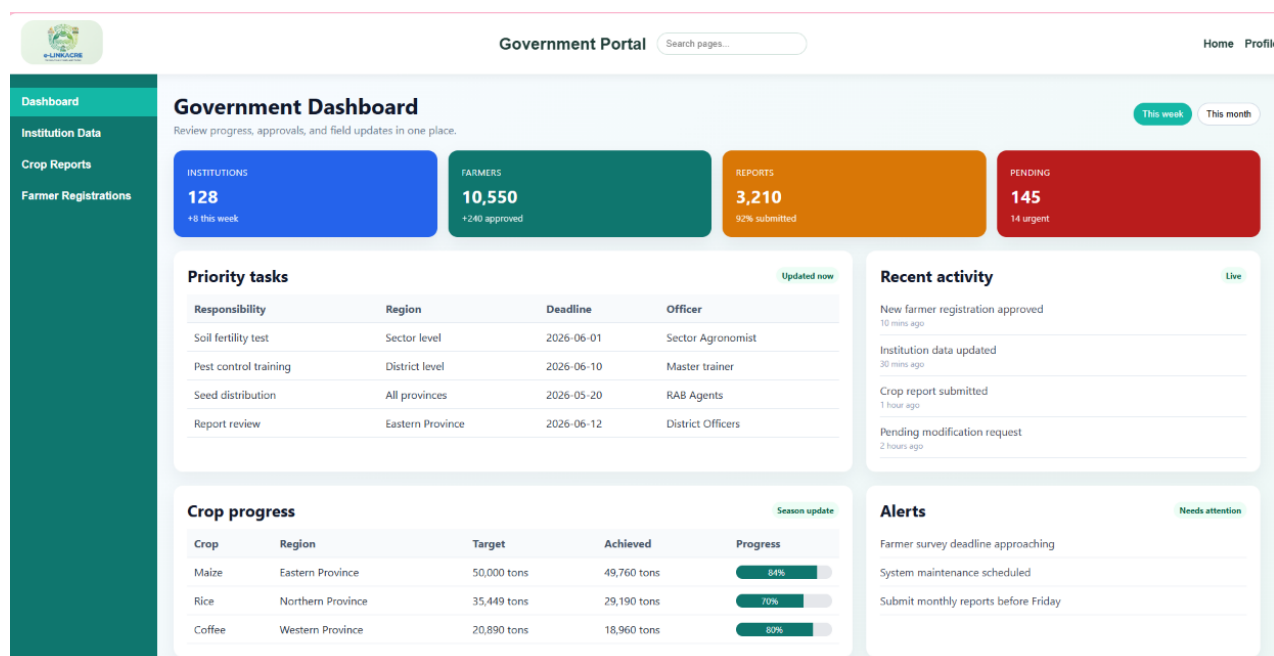
### 3.1. Introduction of the Section

This chapter presents the implemented components of e-LinkAcre and discusses each screen.

### 3.2. Home Page (index.html)



## Government portal screenshot



## Buyer portal screenshot

Buyer Portal

Search products, farms, orders...

Home

Profile

Dashboard  
Marketplace  
My Orders  
Market Trends

## Welcome back, Eulade Traders

You have fresh buying updates today.

NEW PURCHASE

VIEW OFFERS

You can review offers and order updates.

TOTAL SPENDING

**1,420,000 RWF**

+12% from last month

ACTIVE ORDERS

**4 Orders**

2 in transit, 2 pending

PURCHASED QUANTITY

**18.5 Tons**

+3.2 tons this week

AVERAGE CROP PRICE

**320 RWF/Kg**

-4% market trend

### Recent Orders

Updated live

| Order ID  | Crop         | Farmer                 | Quantity | Total Price   | Status     |
|-----------|--------------|------------------------|----------|---------------|------------|
| #ELA-2401 | Wheat        | Jean K. (Musanze)      | 5.0 Tons | 1,600,000 RWF | In-Transit |
| #ELA-2402 | Rice         | Anicet M. (Rusizi)     | 8.0 Tons | 2,800,000 RWF | Completed  |
| #ELA-2403 | Sorghum      | Marie V. (Bugesera)    | 3.5 Tons | 1,050,000 RWF | Pending    |
| #ELA-2404 | Irish Potato | Pierre N. (Burera)     | 2.0 Tons | 600,000 RWF   | Completed  |
| #ELA-2405 | Maize        | Grace T. (Nyagatare)   | 6.5 Tons | 1,950,000 RWF | Pending    |
| #ELA-2406 | Beans        | Fiston R. (Gicumbi)    | 4.2 Tons | 1,260,000 RWF | In-Transit |
| #ELA-2407 | Coffee       | Immaculee D. (Karongi) | 1.8 Tons | 900,000 RWF   | Completed  |

### Recent Activity

Live

- New wheat listing added in **Musanze** (12 Tons available)  
10 mins ago
- Rice purchase from **Bugarama** has been confirmed  
1 hour ago
- Sorghum market prices decreased by **5 RWF/Kg**  
5 hours ago
- Order **#ELA-2402** has been shipped by farmer Anicet  
Yesterday
- Maize demand climbed in **Eastern Province** this morning  
Yesterday
- Invoice for **#ELA-2405** is ready for review  
2 days ago

Market Trends

## Farmer portal screenshot

Farmer Portal

Search pages...

Home

Profile  
Input Portal  
Production  
Reports  
Settings

**Ella K Indebero**

District: Rwamagana  
Farm size: 5 hectares  
Status: Active

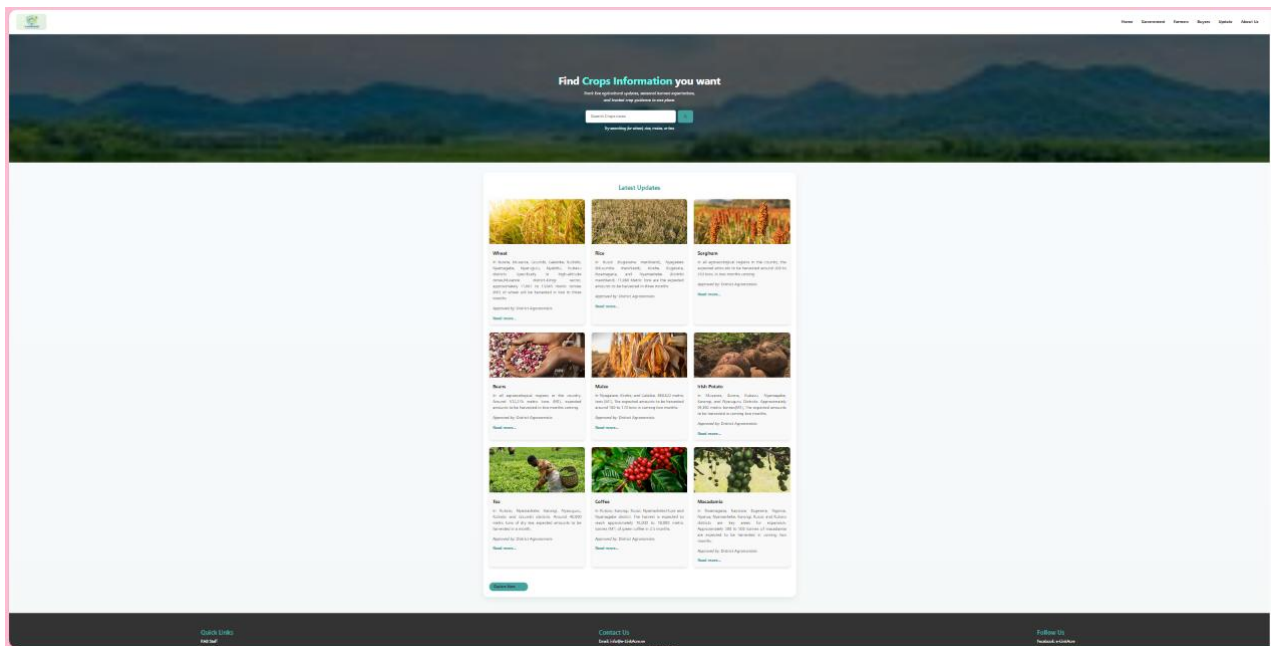
| Item        | Record                 |
|-------------|------------------------|
| Farmer ID   | RW-2026-001            |
| National ID | 1 1987 8 06231 0 32    |
| Phone       | +250 788 123 456       |
| Email       | ellak@example.com      |
| Crops       | Maize, Beans           |
| Cooperative | Kicukiro Farmers Union |

Update profile

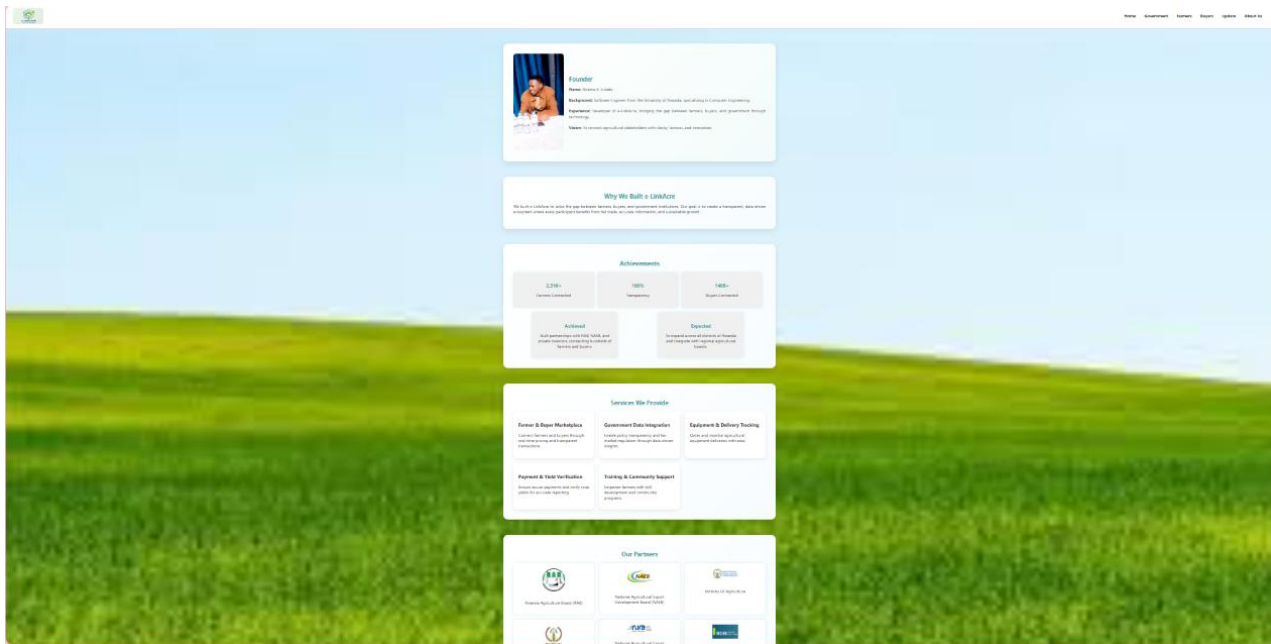
View report

127.0.0.1:5500/pages/production.html

## Update page screenshot



About us screenshot



Landing & shared pages

- **Home (index.html):** Landing page with the hero banner, mission cards, and links into the Farmer, Government, and Buyer portals.
- **About Us :** Founder note, why the platform was built, achievement stats (farmers/buyers connected), and the services offered.
- **Login :** Sign-in form (email/password) plus quick links to policies and contact details.
- **Sign Up :** Account creation form with a user-category selector so a new user registers as farmer, buyer, or government staff.
- **Reset Password :** Email-based password reset request form, with the same quick links and contact footer as login.

- **Navigation** : A standalone header/nav bar component reused across the farmer, government, buyer, and update pages.
- **Updates** : Seasonal harvest listing showing current status per crop (wheat, rice, sorghum, beans, maize, Irish potato, tea, coffee).

#### Farmer portal

- **Farmer Portal (hub)** : Entry page for farmers, linking to Input, Production, Profile, Report, and Settings.
- **Input Portal** : Lets a farmer track material request and payments and request new inputs.
- **Request New Material** : Form for submitting a new input/material request.
- **Production Records** : Tracks the farmer's current planting and harvest progress.
- **Profile** : Farmer's personal details — name, district, farm size, and account status.
- **Report Portal** : Summary view combining profile info, input records, and production records, with farm notes and season totals.
- **Settings** : Preferences for language, district, password, and notifications.

#### Government portal

- **Government Portal (hub)** Entry page for government/agronomist staff, linking to Dashboard, Registration, Crops data, and Institution.
- **Dashboard** : Overview of priority tasks, recent activity, crop progress, and alerts, filterable by week or month.
- **Registered Farmers** : List of farmers with status filters (active, pending, inactive) for review and approval.
- **Crops Info** : Crop reports showing assignment status (approved, pending, scheduled).
- **Institution Data** : Profile of the overseeing ministry/institution and its responsibilities for the season.

#### Buyer portal

- **Buyer Portal (hub)** Entry page for buyers, linking to Dashboard, Marketplace, Orders, and Trends.
- **Buyer Dashboard** : Welcome overview with quick actions (new purchase, view offers) and recent orders.
- **Crops Marketplace** : Browse available harvests with filters for crop type, region, and price, and place orders.
- **My Orders** list with status tabs (all, pending, in transit, completed) and per-order tracking.
- **Market Trends** : Compares direct platform prices against market prices, plus a national crop pricing index.

## Chapter 4. Conclusion and Recommendation

### 4.1. State of the Implemented System

The public home page of e-LinkAcre is implemented and clearly reflects the project's stated vision of connecting Farmers, Government and Buyers, including a visible district-agronomist approval marker on harvest information. The Farmer, Government and Buyer portals, the Updates page, the About Us page and the Login/Sign Up flow are defined in the navigation.

### 4.2. Recommendations

#### **To the Lecturer on Technicalities**

I recommend that lecturer helps me in advance in other website technologies.

#### **To the School, on Time Constraints**

Given the scope of a three-portal platform with an approval workflow and district-level reporting, additional time for backend integration (authentication, database, and the harvest-approval workflow) is recommended beyond the front-end build completed so far.

#### **To the Partners (RAB, NAEB, District Agronomist Offices)**

Early engagement with district agronomists is recommended to validate the proposed approval workflow before the platform is deployed for real use.

#### **To the Users (Farmers, Government Officials, Buyers)**

Basic digital-literacy training and onboarding support are recommended, particularly for farmers and district agronomists who may be using a web-based reporting tool for the first time.

#### **To the Beneficiaries (Farming Communities and Buyers)**

Continued feedback from farming communities and buyers during a pilot phase is recommended to refine pricing transparency and the harvest-update experience.

## Appendix

Link to GitHub: [eulade-rwema/e-LinkAcre](https://github.com/eulade-rwema/e-LinkAcre)

Link to GitHub pages deployment: <https://eulade-rwema.github.io/e-LinkAcre/>